

resistors. There are also 100-kilo-ohm grounding resistors connected to the transistors' gates. The drains (D) are connected to the winding ends of transformer X2. The centre tap of the primary winding is connected to the rectified DC supply from the mains. (This rectified voltage is not filtered; it is just unfiltered, rectified sine wave at point P.)

The power transistors (IRF840) switch the rectified sine voltage supply at the PWM frequency produced by the microcontroller. To smooth out the pulse switching, a 2.5µF, 400V AC fan capacitor is connected across the primary winding of transformer X2. The voltage induced in the secondary winding is a sine wave whose amplitude depends on the width of pulses at points A and B. The program changes the PWM width, and thus the amplitude of the sine wave, to adjust the mains voltage to 220V level.

We thus get a sine wave of the mains frequency. By serially adding the secondary voltage to the input mains voltage we get the stabilised voltage at the output of the unit.

On alternate half cycles, pulses at points A and B arrive to make either of the two transistors T1 and T2 conduct and allow the current to flow through the primary winding. A 2.5µF, 400V AC capacitor is required across the primary winding of transformer X2.

Otherwise, only the pulses from A will pass through and buck and boost cannot be obtained.

In manual position of the switch, when potmeter VR1 is varied from bottom to top (0 to 5V), the voltage across

the secondary decreases, crosses zero and then increases again. This means the secondary voltage varies with the potmeter position. Check the variation in secondary voltage by using a voltmeter, or a multimeter set to 50V AC range. The voltage should increase on either side of the mid-point of VR1. In auto position, combining the secondary output voltage of transformer X2 with the mains voltage gives you the stabilised output.

Testing

1. First, test the controller circuit (Fig. 1) for pulse-width-modulated signals at points A and B. Check changeover from A to B by applying 0V and 5V at point D.

2. Check the circuit for a square wave of 5V amplitude at point D during positive half cycles of AC mains. This square wave is generated by the transistor fed with the unfiltered low voltage DC from transformer X1.

3. Vary the potmeter in manual position of the switch. Using a CRO, you can see variation in the pulse-width (see Fig. 4).

4. As VR1 is adjusted beyond the mid position, pulses at points A and B toggle.

Note that the transistors are 'hot' and 'live.' Energise the voltage stabiliser controller circuit first

but only in manual position of switch S1. Join points C and E and then switch on mains power for the buck-and-boost circuit.

Measure the AC voltage across the secondary output of transformer

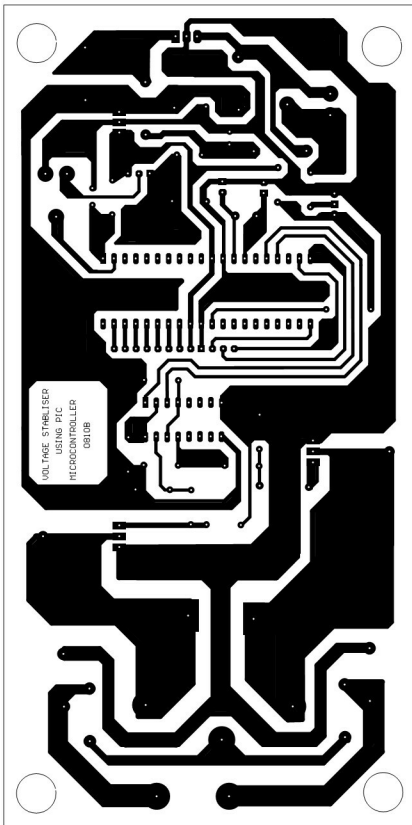


Fig. 6: An actual-size, single-side PCB for the voltage stabiliser using PIC16F877A microcontroller